1

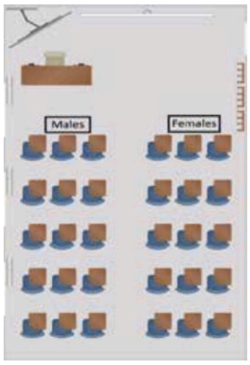
SWE 4101

Introduction to Software Engineering

Lec - 1

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Classroom Rules



- Male and female students will sit in different columns/rows in the classroom. No male and female students can seat side-by-side.
- In case of individual sitting arrangement parted (tablet-chair), they may seat “next” to one another.
- If the number of female students is less than that of the male students, female students may seat in the front rows or front desks of the column. The rest of the seats located in the backside could be filled by the male students

√ Be on time (If you're early, you're on time. If you're on time, you're late)

√ Be a good Listener

√ Ask questions

√ Share new ideas

√ Contribute to discussions

√ No unethical Request

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Google Classroom & Online Class Info

Join using Class Code

jssefg3q

Use only IUT email to join

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Course Objective

Upon completion of this course, you will be able to:

01 Understand basic concepts of computer systems, hardware, software, and data representation.

02 Explain computer organization, networking, and emerging computing technologies.

03 Demonstrate understanding of software engineering principles, processes, and maintenance.

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Course Outcome(CO)

01 Understand the core components of a computer system - hardware, software, and their interaction.

02 Explain basic computer organization, networking, and web technologies with real-world applications.

03 Illustrate key stages of the software life cycle and core software engineering concepts.

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Outcome Based Education (OBE)

Exam Result is Not the Most Important Consideration by Employer

- Communication (verbal & written)
- Honesty/Integrity
- Teamwork skills
- Interpersonal skills
- Strong work ethics
- Motivation & initiative
- Flexibility/adaptability
- Analytical skills
- Computer skills
- Creativity
- Organizational skills
- Detail oriented
- Leadership skills
- Self confidence
- Friendly/outgoing personality
- Well mannered / polite
- Tactfulness
- CGPA
- Sense of humor
- Entrepreneurial skills/risk taker

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Data vs Information

Suppose IUT wants to understand the patterns of students arriving late to classes so that we can take steps to improve punctuality. You are asked to design a system to track late arrivals throughout the semester. What data should be collected about students arriving late? Whainformationcan be generated from that data to help the IUT authority?

Data can be collected:

- √ Student ID
- √ Date
- √ Time of arrival
- √ Course Code
- √ Teacher's name
- √ Reason for being late

Information that can be generated from the data:

- √ Students who are frequently late
- √ Average delay per student or per class
- √ Days or subjects with the highest late arrivals
- √ Patterns over weeks or months (e.g., mornings vs. afternoons)
- √ Most/Least Frequent cause of late arrival

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Core Features of Computers

- √ Speed: Processes data and performs calculations far faster than humans.
- √ Accuracy: Executes tasks with high precision when properly programmed.
- √ Storage: Holds vast amounts of data, from documents to multimedia.
- √ Connectivity:Links to networks and the internet for communication and sharing.
- √ Versatility:Performs a wide range of tasks across different industries.
- √ Automation:Automates repetitive tasks, improving efficiency.

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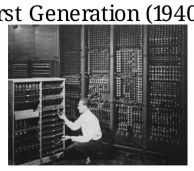
Key Components of Computer



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Generations of Computers

First Generation (1940s–1950s)



- √ Used vacuum tubesand magnetic drums
- √ Large, expensive, and heat-intensive.
- √ Examples:ENIAC, UNIVAC I

Second Generation (1950s–1960s)



- √ Replaced vacuum tubes with transistors → smaller & reliable.
- √ Introduction of assembly language & high-level languages (Fortran, COBOL).

Second Generation (1960s–1970s)



- √ Integrated Circuits (ICs) replaced transistors.
- √ Smaller, faster, more reliable.
- √ Features: Operating systems, multiprocessing

Fourth Generation (1970s–1980s)



- √ Microprocessors → PCs became common.
- √ Innovations: GUIs, software applications, internet access

Fifth Generation (1980s–Present)



- √ Focus on AI, machine learning, quantum computing.
- √ Features: Parallel processing, intuitive interfaces, mobile & connected devices.

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Types of Computers (by size)



Supercomputers:Extremely powerful, perform trillions of calculations/sec for tasks like simulations and data analysis. Examples: IBM Summit, Sunway TaihuLight



Mainframe ComputersLarge, powerful systems for handling bulk data processing, transactions, and critical applications. Examples: IBM zSeries, Unisys ClearPath.



MinicomputersMid-sized computers for small businesses and departmental tasks. Examples: DEC PDP-11, IBM AS/400.



Microcomputers:Personal computers for individual use, like desktops and laptops. Examples: Dell Inspiron, Apple MacBook.



Workstation ComputersHigh-performance PCs for technical or scientific tasks. Examples: HP Z Workstation, Dell Precision.



Embedded Computers:Specialized computers built into devices to perform specific tasks. Examples: Arduino, Raspberry Pi.

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Types of Computers (by Data Type)

